

INTERNSHIP OFFER (MASTER 2)

Shape-changing Interfaces for better communication of the status of machines in technological facilities

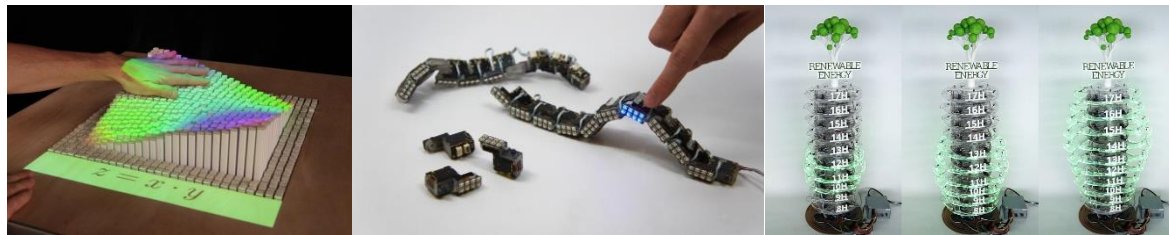


Figure 1 – InFORM [Follmer et al., 2013], ChainFORM [Nakagaki et al., 2016], CairnFORM [Daniel et al., 2021]

Context

This internship is part of the interdisciplinary scientific research project BEST, which aims to find solutions to improve design, manufacturing, and organization methods across various industrial sectors. In this internship, our focus is on technological facilities, physical spaces dedicated to experimenting, researching, and developing future technologies. These facilities serve as meeting places for businesses, researchers, and students to design, manufacture, and test prototypes using machines such as scanners, printers, cutters, drills, milling machines, and electronic stations.

Problem

Today, we observe that crucial information about machines, such as their status, remaining usage time, and planned utilization, is rarely shared with users present in the technological facility, hindering productivity. For instance, when a user wants to switch from one machine to another, they usually have to interrupt their task and approach a facility manager to obtain this information.

Approach

To address this issue, this internship explores shape-changing interfaces (see Figure 1), designed with the assumption that, in the near future, physical objects will become as malleable as virtual objects. These novel interfaces utilize physical shape-change as input and/or output for the computer system. Unlike graphical interfaces, tangible interfaces have the advantage of physically existing in the environment, making them particularly interesting for communicating crucial information about machines at the right place and time in a technological facility.

Internship mission

In this internship, your task will be to assist in the design and manufacturing of a shape-changing tangible interface to communicate crucial information about machines (e.g., status, remaining usage time, planned utilization) to users in a technological facility.

Required qualifications:

- Completed the first year of Master in Robotics, Mechatronics, Computer Science, or Electronics.
- Self-motivated and creative, able to take on initiative.
- Experience with projects mixing mechanics, electronics, and informatics.

Appreciated skills:

- Experience in designing mechanical parts using computer-aided design software (e.g., OnShape, Fusion 360, AutoCAD).
- Experience in designing a printed circuit board using computer-aided design software (e.g., Fritzing, EasyEDA, KiCad).
- Assembly experience with mechanical parts and electronic components (e.g., 3D-printed object integrating a microcontroller, sensor, and motor).
- Programming experience with a microcontroller to read sensors and control actuators (e.g., Arduino, ESP, Feather).

Internship Host Location : [ESTIA-Recherche](#), Bidart (64)

Preferred Duration of the Internship: 6 months

Expected Start Date: From January 1, 2024

Application Procedure: Send your CV and cover letter to m.daniel@estia.fr

Tuteurs du stage :

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